

CIRCUIT 3:

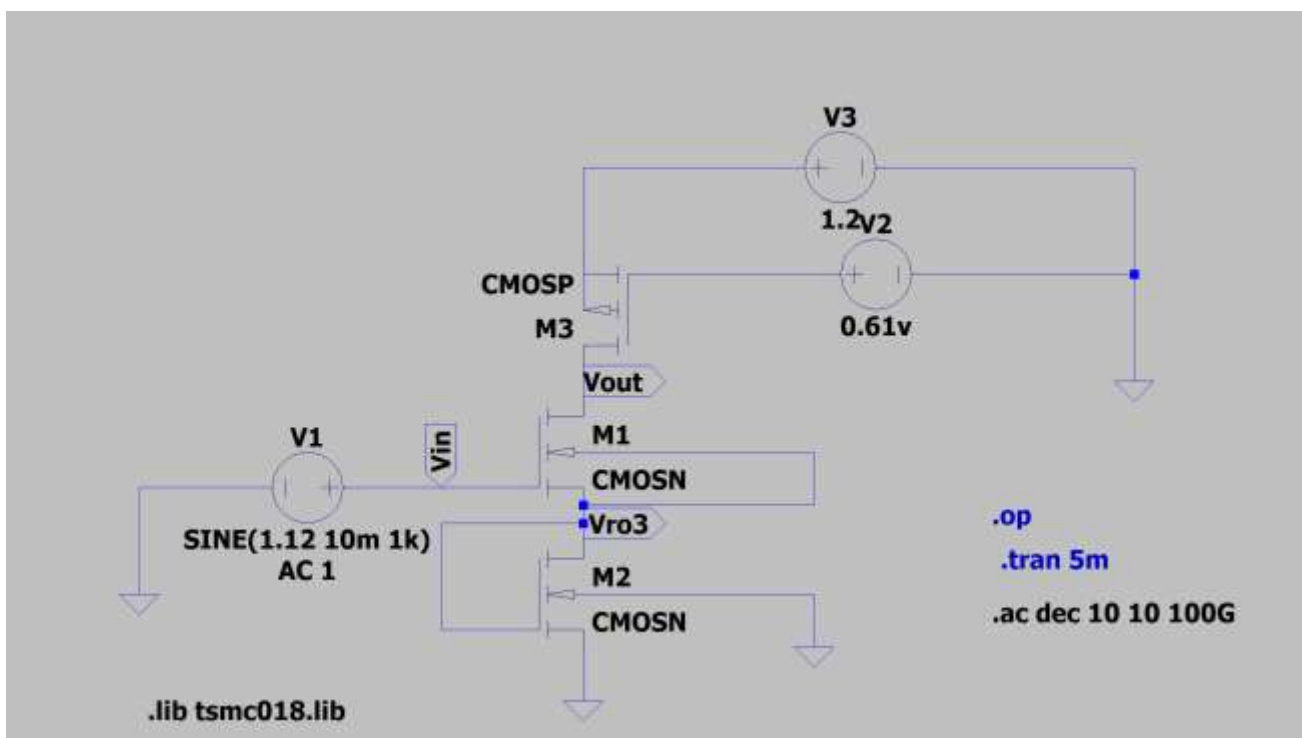
AIM:

To design and simulate three MOSFET amplifier configurations using 180 nm CMOS technology in LTspice, perform DC, transient, and AC analyses, and compare their gain, bandwidth, power consumption, and overall performance.

Given Specifications $V_{DD} = 1.2 \text{ V}$

$$I_D = 200 \mu A$$
$$VOV = 0.2 \text{ V}$$
$$C_L = 0.5 \text{ pF}$$
$$L_n = L_p = 180 \text{ nm } P \leq 0.4 \text{ mW}$$
$$\epsilon_r = 3.9 \quad \epsilon_0 = 8.854 \times 10^{-12} \text{ F/m} \quad t_{ox} = 4.1 \times 10^{-9} \text{ m} \quad \mu_n = 273.809 \text{ cm}^2/\text{Vs} \quad \mu_p = 115.689 \text{ cm}^2/\text{Vs}$$

Circuit diagram:



Calculations:

$$VOV=0.2v$$

$$VOV=V_{GS}-V_{TH} \text{ (NMOS)}$$

$$VOV+V_{TH}=V_{GS}$$

$$0.2+0.36=V_{GS}$$

$$V_{GS}=0.56V$$

$$V_S=0$$

$$V_G=0.56V \text{ (M2)}$$

$$V_G=V_D \text{ (FIXED BIASING)}$$

$$V_{GS}=V_{TH}+VOV$$

$$V_{GS}=V_{TH}+0.2V$$

$$V_{GS}=0.56V$$

$$V_S \text{ OF M1 is equal to } V_D \text{ of M2}$$

$$v_s=0.56v$$

$$v_g=0.56+0.56=1.12v \text{ (M1)}$$

$$V_{SG}=V_{TH}+VOV = 0.39+0.2 = 0.61v$$

$$v_s=v_{dd}=1.2v \quad v_g=0.61v \text{ (M3)}$$

Check Regions of Operation:

M1 (Amplifying NMOS)

$$\text{Source} = V_{ro3} = 0.556 V$$

$$\text{Gate} = V_{in} = 1.12 V$$

$$V_{GS1}=1.12-0.556$$

$$V_{GS1}=0.564V$$

$$VOV1=V_{GS}-V_{TN}$$

$$VOV1=0.564-0.36$$

$$V_{DS1}=0.882-0.556$$

$$V_{DS1}=0.326V$$

$$V_{DS1}>V_{OV1}$$

$$0.326>0.204$$

M1 is in saturation

M2 (Diode Connected NMOS)

$$V_{DS2}=V_{GS2}$$

$$V_{GS2}=0.556V$$

$$V_{GS2}-V_{TN}=0.556-0.36=0.196v$$

$$V_{DS2}=0.556>0.196$$

M2 is in saturation

M3 (PMOS Load)

$$\text{Source} = 1.2V$$

$$\text{Gate} = 0.61V$$

$$V_{SG3}=1.2-0.61$$

$$V_{OV3}=0.59-0.39$$

$$V_{OV3}=0.20V$$

$$\text{Drain} = 0.882V$$

$$V_{SD3}=1.2-0.882=0.318v$$

$$V_{SD3}>V_{OV3}$$

$$0.318>0.20$$

PMOS also in saturation

Maximum Output Voltage

maximum output voltage Upper limit occurs when PMOS M3 leaves saturation.

$$V_{SD3}\geq V_{OVp}$$

$$V_{SD3}=V_{DD}-V_{out}$$

At boundary:

$$V_{out,max}=V_{DD}-V_{OVp}$$

$$V_{out,max}=1.2-0.20=1.0v$$

Minimum output voltage:

Lower limit occurs when NMOS M1 leaves saturation.

Condition:

$$V_{DS1} \geq V_{OV}$$

$$V_{DS1}=V_{out}-V_{ro3}$$

$$V_{out,min}=V_{ro3}+V_{OVn} \text{ (at boundaries)}$$

$$V_{out,min}=0.556+0.20 = 0.756v$$

Check Symmetry

$$V_{out,Q}=0.882V$$

Upper swing:

$$1.0-0.882=0.118V$$

$$\text{lower swing} : 0.882-0.756=0.126V$$

$$I_D = (1/2) \mu_{Cox} (W/L) V_{OV}^2 \text{ Rearranging for } W: W = (2 I_D L) / (\mu_{Cox} V_{OV}^2)$$

$$\text{NMOS Width Calculation } W_n = (2 \times 200 \times 10^{-6} \times 0.18 \times 10^{-6}) / (0.02738 \times 8.42 \times 10^{-3} \times (0.1)^2)$$

$$\text{Numerator: } = 7.2 \times 10^{-11} \text{ Denominator: } = 2.305 \times 10^{-6} \text{ } W_n = 3.12 \times 10^{-5} \text{ m } W_n = 31.2 \mu\text{m}$$

$$\text{NMOS Width } W_n \approx 31 \mu\text{m}$$

(M1 and M3 if identical)

$$\text{PMOS Width Calculation } W_p = (2 \times 200 \times 10^{-6} \times 0.18 \times 10^{-6}) / (0.01157 \times 8.42 \times 10^{-3} \times (0.1)^2)$$

$$\text{Denominator: } = 9.74 \times 10^{-7} \text{ } W_p = 7.39 \times 10^{-5} \text{ m } W_p = 73.9 \mu\text{m}$$

PMOS Width $W_p \approx 74 \mu\text{m}$ if we uplad this value in simulation we will get I_d has 120uA ,to get I_d has 200uA we should increase the wisth

DC ANALYSIS:

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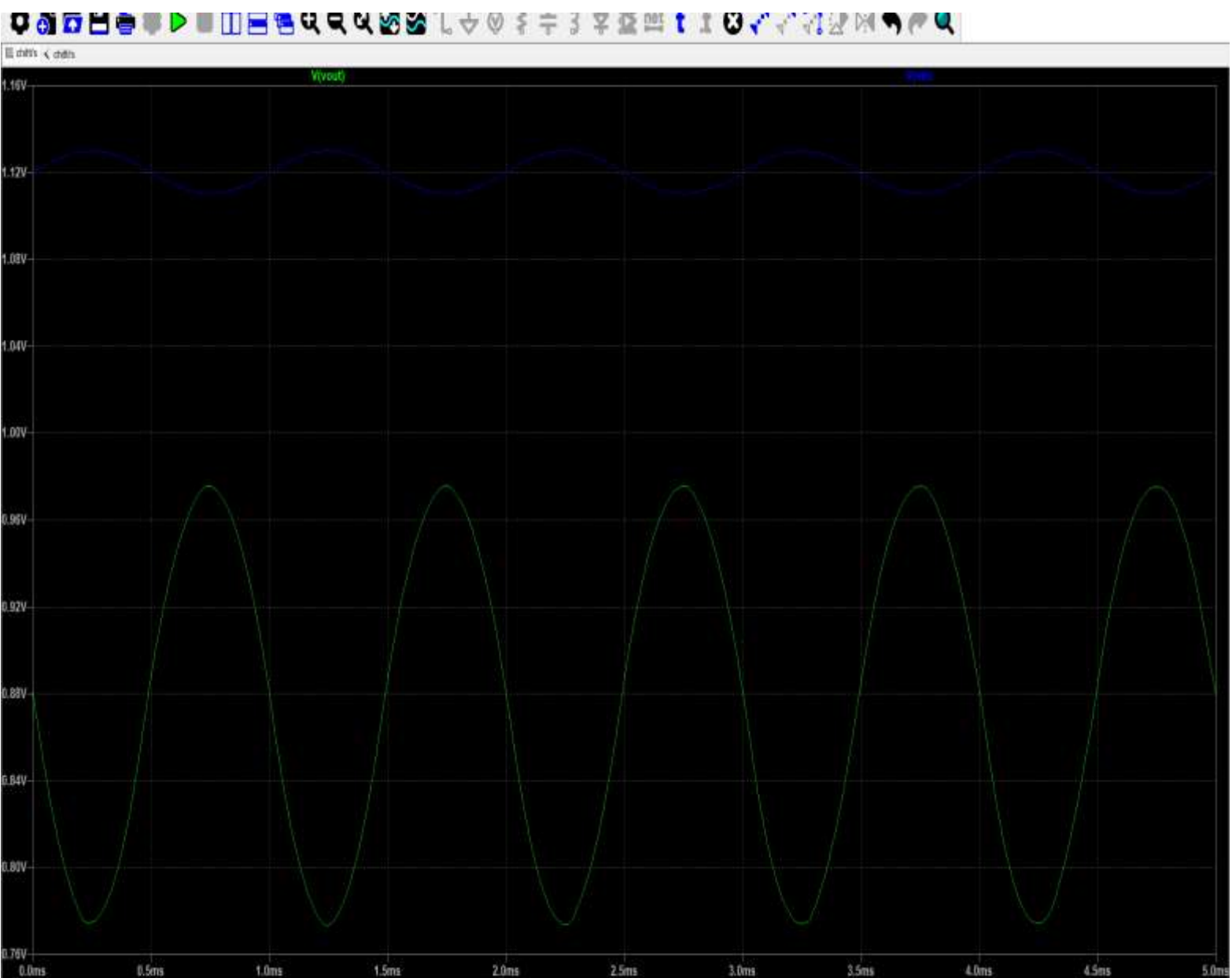
--- Operating Point ---

V(n001):	1.2	voltage
V(n002):	0.61	voltage
V(n003):	1.07686	voltage
V(vin):	1.12	voltage
V(vout):	0.881765	voltage
V(vro3):	0.556361	voltage
I(V1):	0	device_current
I(V2):	0	device_current
I(V3):	-0.000199578	device_current
Ib(M1):	-3.35405e-13	device_current
Ib(M2):	-5.66361e-13	device_current
Ib(M3):	-1.07686e-12	device_current
Id(M1):	0.000199578	device_current
Id(M2):	0.000199578	device_current
Id(M3):	-0.000199578	device_current
Ig(M1):	0	device_current
Ig(M2):	0	device_current
Ig(M3):	-0	device_current
Is(M1):	-0.000199578	device_current
Is(M2):	-0.000199578	device_current
Is(M3):	0.000199578	device_current

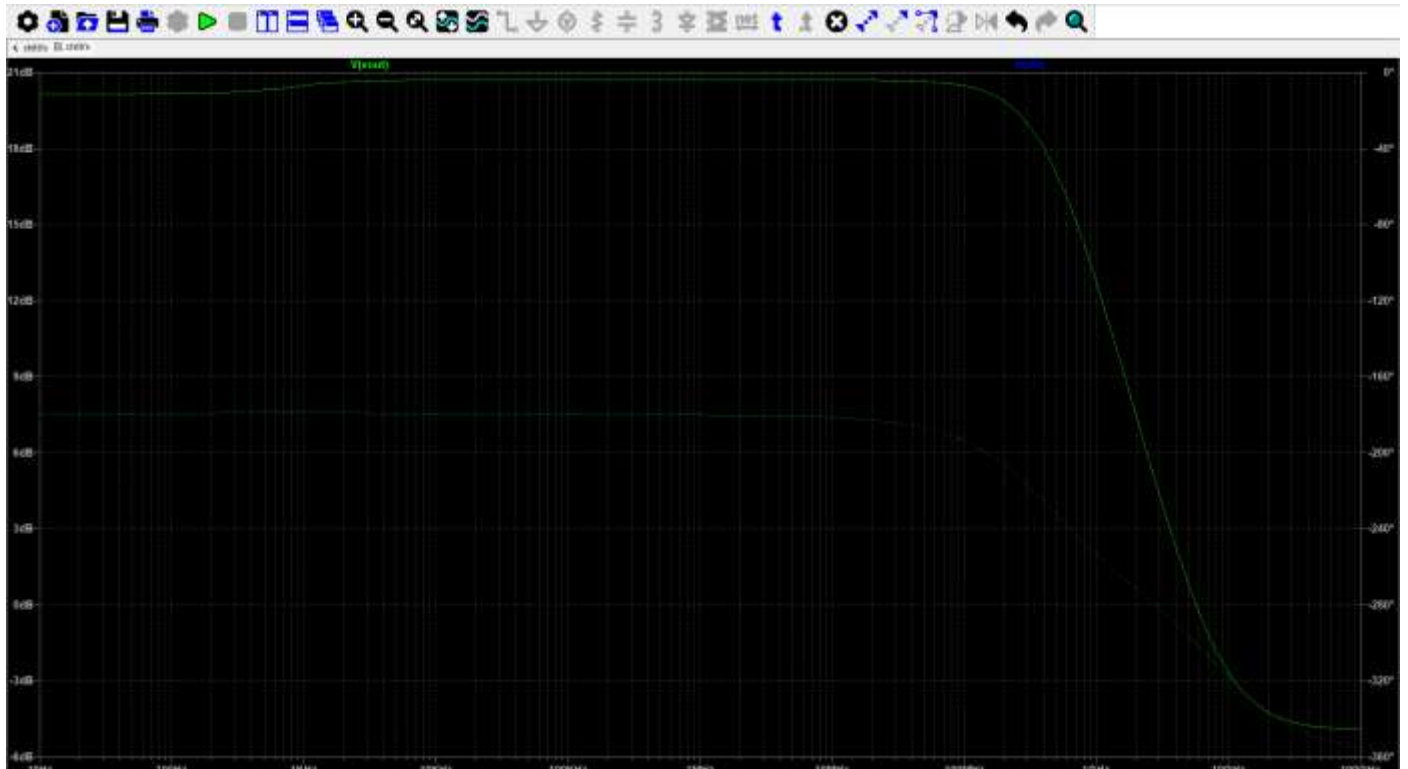
so at $w_p=49.6u$ and $w_n=33.567u$ we will get I_d has approx. $200U_a$

Transient Analysis:

$AV=10.18$ $AV(db)=20*\log(10.18)=20.154$



AC ANALYSIS:



Circuit 3 achieves highest UGB .

Result:

Circuit 3 gives the highest practical midband gain (~18.6 dB).

Diode-connected load reduces gain but simplifies biasing.

Gain-bandwidth trade-off clearly observed across circuits.

circuit validate 180 nm CMOS analog design principles.